
Aim Short to Reach Far: Your Frozen World Model Can Plan Better Than You Think

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Abstract

Visual world models predict action outcomes, while planners commonly score those outcomes by their distance to the encoded goal image. We show that this target can limit control even with exact dynamics and globally optimal short-horizon search: reaching a goal may require actions that initially move away from it. With frozen LeWM models, intermediate targets substantially improve action synthesis and recorded-action ranking on Cube, PushT, Reacher, and TwoRoom. Learned targets and targets drawn from observed experience both produce these gains. We introduce *Anchored Planning*, which retrieves a recorded segment whose start and end resemble the current and goal observations, then aims at an observation shortly after its start. The frozen model scores actions toward this target from the current state. Without additional training, observed targets outperform the released LeWM on every task in our long-range evaluation. Additional final-goal search falls short of the same gains. Target placement and retrieval timing further shape success, while lower successor-prediction error need not translate into better control. Changing only the target lets the same frozen model and planner reach goals that final-goal scoring misses.

1 Introduction

Visual world models predict how observations change under candidate actions [Zhou et al., 2025, Assran et al., 2025, Maes et al., 2026]. A common planner selects the predicted endpoint closest to the encoded goal image, executes a short block, and replans [Ebert et al., 2018, Hafner et al., 2019]. This couples two choices that objective-driven architectures assign to separate modules [LeCun, 2022]: how to predict an action’s effect and which outcome to pursue. We show that changing only the target lets the same frozen predictor and planner reach goals that final-goal scoring misses.

Reaching a goal can require first moving farther from it. Classical potential-field methods encounter this difficulty when local descent leads into a nonglobal minimum [Khatib, 1986, Rimon and Koditschek, 1992]. Short-horizon distance scoring inherits the same *objective mismatch*: it can reject the first steps of a useful route before seeing where they lead. Proposition 1 makes the distinction precise. In a smooth control problem, exact predictions and globally optimal short-horizon search still select inaction forever, whereas successive observed targets on the same feasible route lead to the goal. Prediction and search can therefore be correct while the objective favors the wrong action.

With frozen LeWM predictors, learned and observed intermediate targets both substantially improve action synthesis and recorded-action selection. Observed targets have a larger mean advantage over final-goal scoring at longer goal offsets. At the 100–150-action offsets, short observed-target searches also exceed the released LeWM on every task. Additional final-goal CEM iterations fall short of the same gains. The same frozen dynamics support much stronger control once the planner scores progress toward a useful destination (Figure 1).

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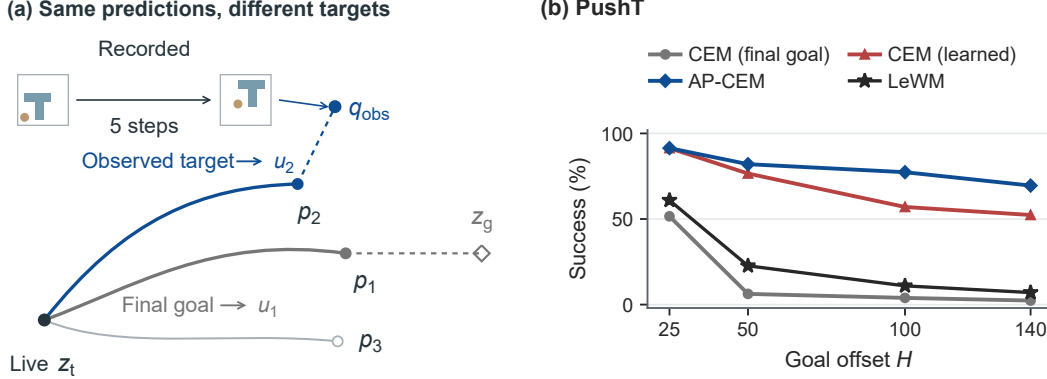


Figure 1: **Changing the target improves control with the same frozen predictor.** (a) Both target rules score the same endpoints $p_i = F(z_t, u_i)$. Final-goal scoring selects u_1 ; observed-target scoring selects u_2 . Solid curves show predictions; dashed segments show scoring distances. The target inset shows a recorded successor; positions and observations are schematic. (b) On PushT, intermediate targets sustain control as the recorded goal offset grows. Curves use paired standard-start queries and the same frozen LeWM weights.

Where should that destination come from? Learned terminal values guide short searches [Hansen et al., 2022, 2024], while high-level world models and subgoal generators supply intermediate targets [Zhang et al., 2026, Cheng et al., 2026]. Recorded trajectories already contain observations encountered on a route to a later endpoint. *Anchored Planning (AP)* retrieves a segment whose start and distant endpoint resemble the current and goal images, then takes an early observed successor as its target. Retrieval looks along the longer recorded route; the frozen model predicts short action blocks from the live state. This separation lets the same target guide action synthesis from observation-only memory or selection from action-labeled memory.

Predicting the recorded successor more accurately does not always produce a better planning target. A target model trained on the same memory estimates held-out successors more accurately on every task, yet observed targets achieve higher mean success in action synthesis. Local placement and retrieval timing further determine which recorded destinations help the predictor guide action.

Our contributions are:

- i) We identify a planning-objective bottleneck: final-goal scoring can stall with exact predictions and optimal short-horizon search, and target changes recover control from frozen visual predictors.
- ii) We introduce Anchored Planning, which uses a recorded route’s early successor as a shared target for action synthesis and ranking, without training another model.
- iii) We show that target choice matters beyond search effort and successor-prediction accuracy, and establish how target placement, retrieval timing, and live-state prediction shape control.

Planning targets should therefore be evaluated alongside prediction accuracy when assessing what a world model can control.

2 Why the Planning Target Matters

A frozen encoder E maps current and goal images to $z_t = E(o_t)$ and $z_g = E(o_g)$. An action block u contains $L = 5$ successive primitive actions. The frozen predictor $F(z_t, u)$ estimates the encoded state after that block and includes the fixed action normalization. The planner seeks the block whose predicted endpoint is closest to target q :

$$\hat{u}(q) \in \arg \min_{u \in \mathcal{C}} J_q(u), \quad J_q(u) = \|F(z_t, u) - q\|_2^2. \quad (1)$$

For a finite candidate set, we score every block; the cross-entropy method (CEM) [de Boer et al., 2005] approximates the continuous-space optimization. The controller executes the selected block and observes again. Physical success is always measured against the final goal. The scoring target q is used to select the next action block.

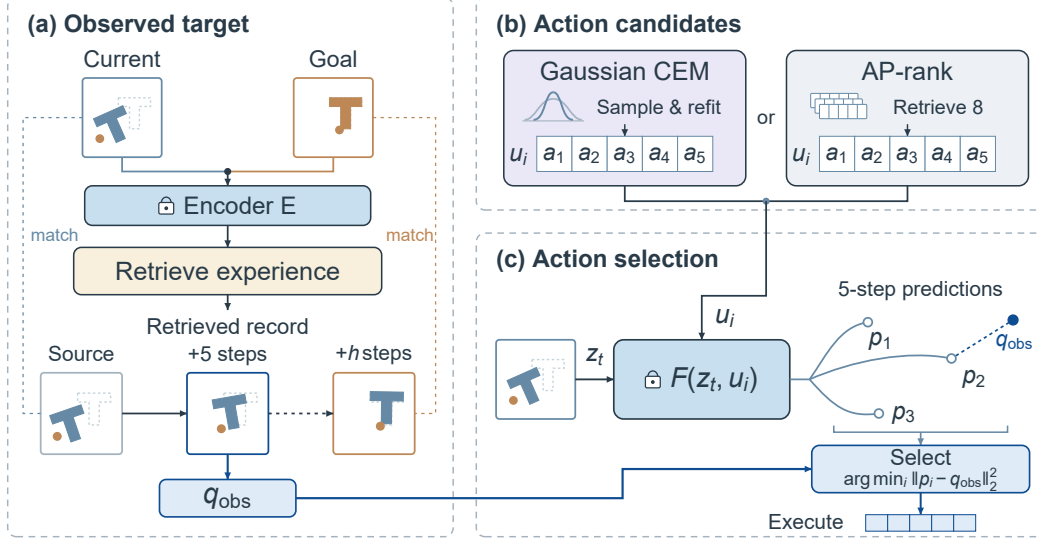


Figure 2: **Anchored Planning: observed targets, live predictions.** (a) Match the current and goal images to recorded source and distant images. The leading record’s five-step successor supplies q_{obs} . The distant offset is $h = \max(5, H - t)$, for goal offset H and t executed actions. (b) Each strip is a candidate block u_i : Gaussian CEM synthesizes actions; AP-rank retrieves eight blocks. (c) The frozen model predicts endpoints $p_i = F(z_t, u_i)$. The dashed segment shows distance to q_{obs} . Select the closest endpoint, execute up to five actions, observe, and repeat. All glyphs and positions are schematic.

For two fixed predictions $p_u = F(z_t, u)$ and $p_v = F(z_t, v)$, let $\Delta_q(u, v) = J_q(u) - J_q(v)$. A negative value favors u over v . Changing the target from q to q' gives

$$\Delta_{q'}(u, v) - \Delta_q(u, v) = 2(p_u - p_v)^\top (q - q'). \quad (2)$$

Thus the same predicted endpoints can favor different actions when the target changes. The following construction shows how this changes control even when the short-horizon search finds the global minimum.

Proposition 1 (Goal-distance control can stall). *Even with smooth dynamics, bounded continuous actions and exact predictions, globally optimal five-action final-goal scoring can select the zero block forever from a continuum of goal-reachable states. Under identical dynamics, actions and execution, six successive endpoints of one feasible recorded trajectory instead lead every such start to the goal.*

The construction places states on the curve $c(s) = (s, s^2)$, with goal $c(2)$. Each primitive action increases s by an amount in $[0, 1/8]$, so exact predictions follow the same continuous route toward the goal. Yet squared final-goal distance along that route is

$$D(s) = \|c(s) - c(2)\|^2 = (s - 2)^2 + (s^2 - 4)^2. \quad (3)$$

For starts $s \in [-3/2, -1]$, every nonzero admissible block ends farther from the goal under D , whereas the zero block leaves the distance unchanged. The global optimum of the local objective is therefore to stay still. The predictor correctly places every nonzero block farther along the route, but the distance score favors staying still.

Replacing the final goal with successive targets on the recorded curve changes this choice. Each target can be attained within the local action horizon in this construction, and selecting it advances along the same dynamics until the goal is reached (see Appendix B for the complete construction and proof). Anchored Planning applies this idea by retrieving a task-relevant trajectory and predicting actions toward an early observed successor. Section 4 measures the resulting target effect with frozen visual predictors.

3 Anchored Planning

Anchored Planning separates choosing a destination from choosing an action: recorded experience supplies an observed target, and the frozen model predicts how candidate actions approach it from the live state (Figure 2). We consider two memory settings. *Observation-only memory* contains

ordered images with episode and time indices; the planner must synthesize actions. *Action-labeled memory* also contains the recorded actions, which the planner can rank. Both use the same frozen predictor, target rule, and fixed memory.

3.1 Use distant experience to choose an observed target

A short search needs a nearby destination that belongs to a route toward the final goal. We therefore match a record through its source and distant observations, then use its early successor to score actions. Each record contains a source, an L -step successor, and an h -step endpoint: retrieval looks along the longer route while prediction covers the next action block.

Each query provides the recorded action offset H between its starting image and goal image. After t executed primitive actions, we retrieve records spanning

$$h(t) = \max\{L, H - t\}. \quad (4)$$

As execution advances, retrieval matches progressively shorter recorded spans to the final goal, down to L . The observed target and prediction horizon remain L steps ahead of their respective source states. This updates the route used to choose a nearby destination.

Let s index a recorded start. For retrieval horizon h , memory $\mathcal{B}_h$ contains starts for which both $s + L$ and $s + h$ remain within the same episode. Each record pairs the retrieval key

$$b_s^{(h)} = (z_s, z_{s+h}, z_{s+h} - z_s)$$

with its successor observation z_{s+L} . Retrieval uses ordered observations and episode indices. For AP-rank, a record additionally provides its next-five-action block $u_s = (a_s, \dots, a_{s+L-1})$. The current query is $b_t = (z_t, z_g, z_g - z_t)$. Retrieval ranks records by standardized squared distance,

$$d_h(t, s) = \left\| \frac{b_t - b_s^{(h)}}{\sigma_h} \right\|_2^2. \quad (5)$$

Here σ_h is the coordinate-wise population standard deviation over eligible records. The closest start s_1 supplies the target

$$q_{\text{obs}} = z_{s_1+L} = E(o_{s_1+L}). \quad (6)$$

3.2 Synthesize actions from observation-only memory

AP-CEM refines a Gaussian over five-action blocks. From a fixed prior, it projects proposals onto the action bounds and scores their predicted endpoints against q_{obs} . The best-scoring proposals refine the distribution. We compare the projected means before and after optimization and execute the lower-cost block. Appendix A gives the complete update and action mapping.

3.3 Rank retrieved actions from the current observation

AP-rank takes one action block from each of the eight retrieved records and predicts its endpoint from the current observation. All eight endpoints are compared with the same target: the closest record’s observed target. Let P clip each action coordinate to its task-specific bounds. The clipped candidate set is $\mathcal{C} = \{P(u_{s_1}), \dots, P(u_{s_8})\}$, and AP-rank selects

$$\hat{u} = \arg \min_{u \in \mathcal{C}} \|F(z_t, u) - q_{\text{obs}}\|_2^2, \quad (7)$$

with ties resolved by retrieval order. The eight predictions share one batch. Execution checks success, termination, and budget after each primitive, then observes and replans after at most five actions (Algorithm 1 in Appendix A.6).

3.4 Target and prediction controls

We compare final-goal, learned, and observed targets under each action rule. Final-goal scoring sets $q = z_g$. A learned-target MLP predicts the observed successor from the current and goal encodings, their difference, and the retrieval span. It trains on the same memory observations used by AP, with its configuration selected on held-out memory episodes (Appendix C.1).

For CEM, the target determines which predicted outcomes refine the Gaussian; initialization, seeds, search settings, and action mapping stay fixed. Ranking scores the same eight retrieved blocks from a shared live state. Each controller then repeats its target rule and action selection along the resulting rollout.

Table 1: **Intermediate targets improve control with the same frozen model.** Success (%) on 128 paired queries per task. Groups distinguish available experience; learned targets train on memory observations. Perturbed averages starts within query; means weight tasks equally. Bold marks the maximum within each multi-controller group. AP-CEM and AP-rank use observed targets.

Task	Start	No memory	Observation-only			Recorded actions			
		LeWM	CEM final	CEM learned	AP-CEM	Direct	Rank final	Rank learned	AP-rank
Cube	Standard	15.6	7.0	62.5	46.9	82.0	62.5	91.4	73.4
	Perturbed	12.5	2.7	38.7	29.7	47.7	41.8	50.0	45.7
PushT	Standard	7.0	2.3	52.3	69.5	51.6	25.8	54.7	66.4
	Perturbed	9.4	2.7	48.8	64.8	18.0	12.9	35.5	47.7
Reacher	Standard	66.4	25.8	54.7	75.8	99.2	90.6	95.3	96.9
	Perturbed	68.8	22.7	59.0	78.9	96.9	87.5	96.9	95.7
TwoRoom	Standard	17.2	1.6	53.1	46.9	99.2	89.1	96.9	98.4
	Perturbed	16.8	1.6	55.1	46.9	100.0	88.7	98.4	100.0
Mean	Standard	26.6	9.2	55.7	59.8	83.0	67.0	84.6	83.8
	Perturbed	26.9	7.4	50.4	55.1	65.6	57.7	70.2	72.3

Direct executes the leading record’s projected action block, then observes and retrieves again. Comparing Direct with predictive ranking measures the benefit of evaluating recorded actions from the current state.

4 Experiments

The experiments trace target choice from its effect on control to the design of useful destinations:

RQ1: How much control does changing the target reveal?

RQ2: Can longer lookahead or more search recover the same gains?

RQ3: What makes a target effective, and how does prediction help reach it?

4.1 Evaluation setup

Models, memory, and paired queries. We evaluate Cube, PushT, Reacher, and TwoRoom with task-specific pretrained LeWM models [Maes et al., 2026]. Encoders, predictors, and original action scalers remain frozen. Memory contains the supplied training trajectories. Each task has 128 query episodes, excluded in full from memory and the prefix pool. A query pairs its start with the image H actions later; these original offsets span 100–150 actions (Table 5). All experiments below use these paired queries. The appendix reports independent-query replications.

Starts and success. Recorded actions can have different effects from a displaced state. We therefore evaluate each recorded (standard) start and two starts perturbed by preassigned five-action prefixes, keeping the goal, memory, and subsequent allowance fixed. Controllers receive images, H , and the allowance. Physical states restore paired starts and measure success after every primitive. Perturbed rates average starts within query; task means weight tasks equally (Appendix A).

Action rules and targets. Single-block CEM uses 30 iterations, 300 candidates, and 30 elites; ranking selects among eight retrieved blocks. Both use final, learned, or observed targets (Section 3.4), replan after at most five actions, and project proposals onto the raw-action Box before prediction and execution. LeWM retains its released action interface and 25-action planning/execution blocks; it matches the released evaluator on shared queries (Appendix D.2).

4.2 RQ1: How much control does changing the target reveal?

Both intermediate targets improve control. Changing the target while retaining the frozen predictor and action rule improves success on every task at standard and perturbed starts (Table 1). At standard starts, CEM’s mean rises from 9.2% with the final goal to 55.7% with a learned target and 59.8% with an observed target. Ranking rises from 67.0% to 84.6% and 83.8%, respectively. Thus both target sources reveal control that final-goal scoring misses; AP obtains its target directly from recorded experience.

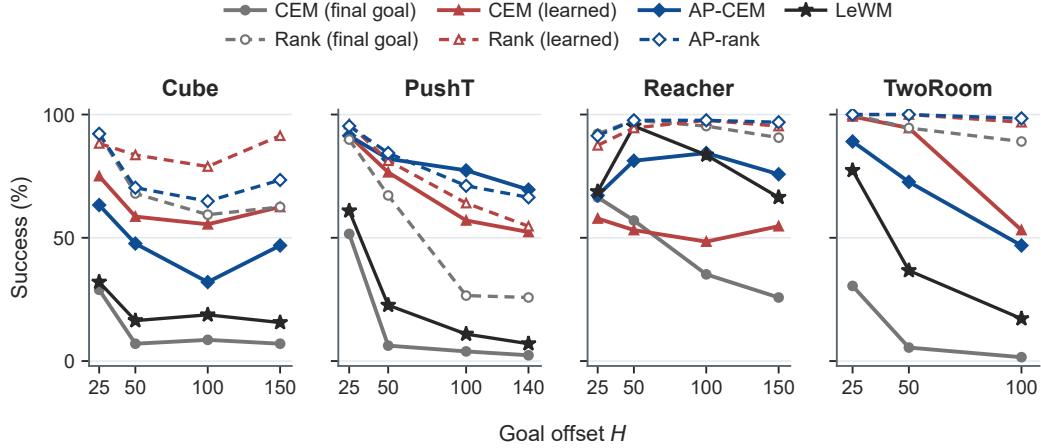


Figure 3: **Mean observed-target gains are larger beyond the shortest offset.** Standard-start queries retain their source as H changes. Action allowance preserves each task’s B/H ; planner lookahead stays fixed. Original-offset points reuse Table 1. Table 12 gives perturbed results.

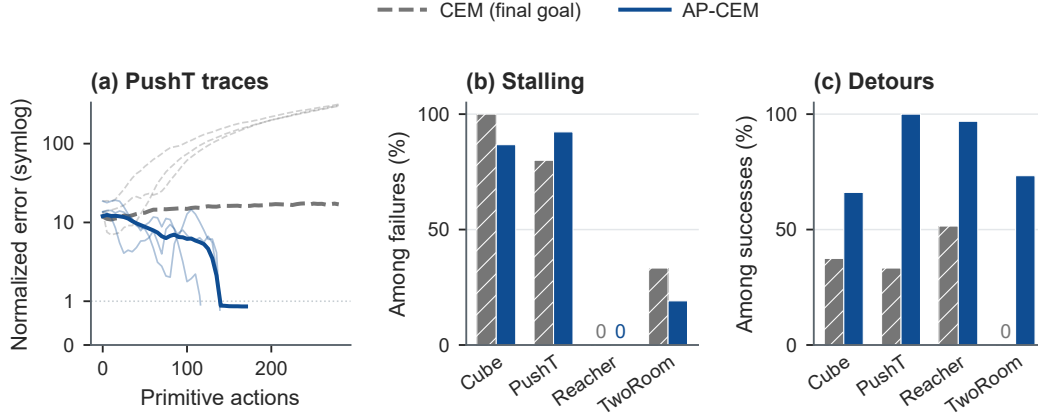


Figure 4: **Successful PushT control takes detours while final-goal failures often stall.** (a) Final-goal CEM failures and AP-CEM successes at standard starts. Thin curves show the first three eligible queries; thick curves show group medians with terminal error carried forward. The axis is linear below one and logarithmic above. (b) Stalling among failures. (c) Detours among successes. Rates cover rollouts that enter control; Appendix C.2 gives counts and threshold sensitivity.

Target choice matters more at longer offsets. We vary goal offsets with starts and lookahead fixed (Figure 3). Final-goal CEM declines from the shortest to the original offset on all tasks. Under their PushT protocol, Caselli et al. [2026] report flat LeWM success of 94.0%, 52.7%, and 18.0% at offsets 25, 50, and 75. From $H = 25$ to $H = 100$, observed targets’ standard-start gains grow from 33.4 to 47.9 pp for CEM and from 1.2 to 15.4 pp for ranking; perturbed starts follow the same trend. Final-goal ranking handles nearby goals; intermediate targets help at longer offsets.

The recovered control includes physical detours. Successful actions can initially increase physical error, just as the construction in Section 2 illustrates. On standard-start PushT, all 89 AP-CEM successes take such a detour, whereas 100 of 125 final-goal CEM failures stall (Figure 4). Error is normalized by the task’s success thresholds; detours raise it above a previous minimum, and stalling denotes little task-relevant motion near a rollout’s end.

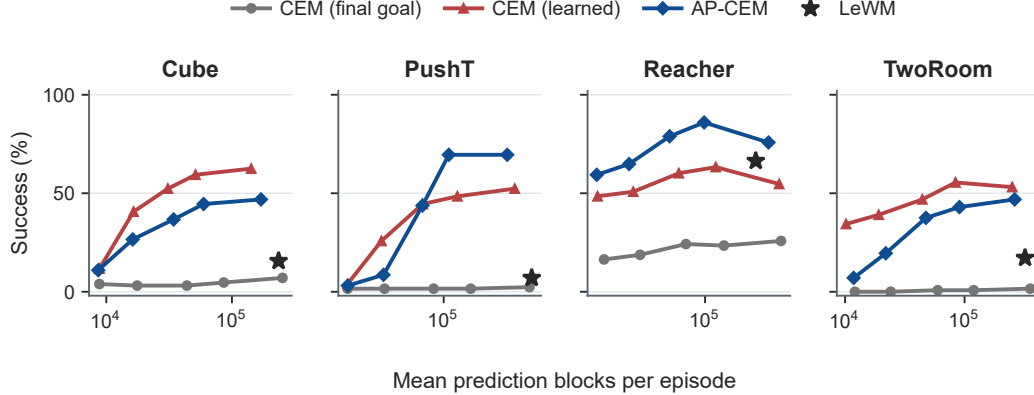


Figure 5: **Two intermediate-target iterations exceed 30 final-goal iterations on every task.** Standard-start curves join iteration counts 1, 2, 5, 10, 30; stars mark LeWM. Work counts mean predicted five-action blocks per episode, including early stopping. The same comparison holds at perturbed starts (Table 9).

Table 2: **Lower successor error does not always mean higher control success.** Mean squared distance to the recorded successor uses held-out memory transitions; CEM success uses standard-start evaluation queries. Bold marks higher success. Appendix C.1 describes the validation measurements.

Task	Successor error		CEM success (%)	
	Learned	Observed	Learned	Observed
Cube	19.708	63.056	62.5	46.9
PushT	3.774	10.668	52.3	69.5
Reacher	74.724	154.351	54.7	75.8
TwoRoom	133.857	266.676	53.1	46.9

4.3 RQ2: Can more planning recover the same gains?

Target changes outperform more final-goal search. RQ1 held search fixed; here we increase its budget and lookahead. With just two CEM iterations, both intermediate targets exceed 30-iteration final-goal CEM on every task at both starts (Figure 5).

Short observed-target search exceeds LeWM at the original offsets. AP-CEM leads on all tasks at both starts, using 63.4–85.5% of LeWM’s mean episode prediction blocks on Cube, PushT, and TwoRoom, and 126.6–128.1% on Reacher. LeWM leads at some shorter Reacher offsets (Figure 3). AP-CEM also exceeds five-action LeWM replanning across tasks and starts (Table 11).

4.4 RQ3: What makes a target useful for control?

Successor accuracy and control favor different target choices. With the gains from intermediate targets established, we first compare target accuracy with control success. The learned target predicts held-out successors with lower mean error in every task, while observed targets achieve higher mean CEM success (Table 2). Cube and TwoRoom favor learned targets; PushT and Reacher favor observed targets. Lower successor error therefore does not consistently predict higher control success.

Nearby targets and retrieval timing shape success. With five-action predictions, a three-step target raises AP-CEM’s mean success to 69.9%, while AP-rank performs best at the default five-step offset among those tested (Table 3). One- and ten-step targets give lower means than the default for both rules. Shortening the retrieval span during execution is especially consequential for ranking: fixing it at H reduces standard-start success from 83.8% to 9.6%. With the adaptive span, ten percent of memory retains most success, and a key using the current state and goal-conditioned displacement achieves the highest ranking mean.

Table 3: **Target placement and retrieval timing determine how memory guides control.** Success (%) averages the four tasks. Prediction spans $L = 5$ actions; ℓ changes the target’s recorded offset. Memory subsets retain the default target offset. Fixed span uses $h = H$ throughout execution. Bold marks the highest mean per column within each panel.

(a) Target offset and memory size

Variant	AP-rank		AP-CEM	
	Standard	Perturbed	Standard	Perturbed
Target $\ell = 1$	69.5	61.3	43.0	37.9
Target $\ell = 3$	78.5	68.5	69.9	62.6
Target $\ell = 5$ (default)	83.8	72.3	59.8	55.1
Target $\ell = 10$	81.6	68.4	54.1	52.7
Memory 10%	77.5	68.9	56.6	52.1
Memory 25%	80.1	70.4	58.4	53.5
Memory 50%	82.0	71.0	58.6	53.8

(b) Retrieval design for AP-rank

Variant	Standard	Perturbed
Default	83.8	72.3
Key without far endpoint	85.7	72.9
Key without displacement	83.0	72.1
Fixed retrieval span	9.6	11.3

Table 4: **Live-state prediction improves endpoint accuracy and action selection.** Means use task-specific squared latent distances (lower is better); bold marks each metric’s minimum. Displ. transfers recorded motion; Direct selects the leading record. Starts are excluded if any candidate ends before five actions (at most 3 of 128 standard and 6 of 256 perturbed starts per task); counts are in our repository’s `diagnostics.csv`.

Task	Start	Endpoint error		Selection regret		
		Predictor	Displ.	Predictor	Displ.	Direct
Cube	Standard	7.538	35.344	5.332	7.535	11.375
	Perturbed	37.726	66.860	8.472	11.731	14.903
PushT	Standard	1.532	3.078	0.200	0.580	0.849
	Perturbed	2.980	9.562	0.753	2.418	4.449
Reacher	Standard	25.243	51.917	8.644	15.058	18.640
	Perturbed	25.587	50.862	7.401	12.558	16.296
TwoRoom	Standard	13.560	162.035	1.433	29.838	41.749
	Perturbed	12.532	160.497	1.283	20.154	37.307

An observed destination adjusts the requested motion. For recorded displacement $d = z_{s+L} - z_s$ and live offset $e = z_t - z_s$, the transported target $z_t + d$ requests the full recorded motion, whereas the observed target requests $d - e$. Anchoring raises mean CEM success from 53.7% to 59.8% at standard starts and from 46.0% to 55.1% after perturbation. Cube, PushT, and TwoRoom favor anchoring; Reacher favors transport (Table 16).

Live prediction improves the choice of action. To assess how actions approach the target, we execute all eight candidates from each saved simulator start and compare the resulting endpoints with

$$\tilde{p}_j = z_t + (z_{s_j+L} - z_{s_j}), \quad p_j = F(z_t, u_j). \quad (8)$$

The predictor reduces both endpoint error and selection regret relative to displacement in every task and start setting, and reduces regret relative to Direct (Table 4). Error is squared distance to the simulated endpoint; regret is the chosen action’s exact observed-target cost above the best candidate’s cost. Appendix E.3 gives first-block interventions with a common continuation.

This action evaluation has its largest repeated-control benefit over Direct on PushT: AP-rank raises success from 51.6% to 66.4% at standard starts and from 18.0% to 47.7% after perturbation. Experience specifies where to aim, and live prediction determines which recorded action approaches it from the current state.

5 Related Work

Planning objectives. Artificial potential fields turn goals and obstacles into control objectives [Khatib, 1986]; navigation functions use known free-space geometry to remove nonglobal minima [Rimon and Koditschek, 1992]. Latent planners score predicted features [Sobal et al., 2025], while Universal Planning Networks and Monotone Cost Ranking learn representations or costs for control [Srinivas et al., 2018, Chahe et al., 2026]. Terver et al. [2026] study optimizers and latent costs, showing that accurate prediction need not yield successful planning. Concurrent studies test decoded-position and learned temporal costs on TwoRoom [Singh, 2026] or learn costs over latent rollouts [Huang et al., 2026]. We isolate target-induced stalling with exact dynamics and optimal short-horizon search, then compare targets with a fixed learned predictor, metric, and search.

Recorded and generated intermediate targets. SPTM and SoRB search graphs of observations using learned connectivity or distances [Savinov et al., 2018, Eysenbach et al., 2019]; PALMER stitches recorded segments using learned reachability [Beker et al., 2022]. HWM predicts sub-goals with a high-level world model [Zhang et al., 2026], and Hi-LeWM improves their quality by constraining high-level search near encoded training sequences [Caselli et al., 2026]. SAGE learns target and action generators with frozen LeWM, using expert trajectories and low-dimensional state [Cheng et al., 2026]. Its generator-only variant is closest to our observation-only CEM: Gaussian search scores candidates against a learned target. AP reads an observed successor from a retrieved segment as its target. Our learned-target baseline compares successor prediction with retrieval using the same visual inputs and memory.

Experience for action and target choice. VINN predicts actions by local regression on visual neighbors [Pari et al., 2022]; Direct tests immediate action reuse under our current-goal retrieval rule. Experience also supports transition estimation [Malato and Hautamäki, 2025], hybrid LeWM planning [Gao et al., 2026], action or trajectory learning [Florence et al., 2022, Chi et al., 2023, Janner et al., 2022], hindsight goals [Andrychowicz et al., 2017], video rewards [Chen et al., 2021], and offline goal-conditioned benchmarks [Park et al., 2025]. AP gives experience a role in choosing the destination for synthesized or retrieved actions; live-state prediction then evaluates which action approaches it.

6 Conclusion

Changing the target can let the same frozen predictor and planner reach goals that final-goal scoring misses. Our construction shows that final-goal scoring can fail even with exact predictions and globally optimal short-horizon search. On the LeWM tasks, learned and observed intermediate targets achieve higher success than additional final-goal search. Goal-distance planning success, a common world-model benchmark, thus reflects the objective as well as the predictor. Target placement, retrieval timing, and live-state prediction further shape control. Testing the same target choices with other world-model families could show how the gains depend on the predictor.

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The appendix follows the main questions: evaluation protocol and proofs first, then target comparisons and behavior (RQ1), search and goal distance (RQ2), and target construction and action selection (RQ3). The final section reports an independent-query replication. Our repository provides code and machine-readable results.

A Evaluation Protocol

Table 5: **Evaluation inputs.** H is the recorded goal offset, B the primitive-action allowance, and memory the number of eligible episodes.

Task	Cube	PushT	Reacher	TwoRoom
H	150	140	150	100
B	150	280	300	200
Memory	8,000	16,688	8,000	8,000

A.1 Query construction and retrieval memory

Memory consists of the offline training trajectories provided with the task models. LeWM permits both pseudo-expert and exploratory behavior-policy data [Maes et al., 2026]. Ordered observations, temporal indices, and episode boundaries are available in both memory settings; recorded action values are additionally available to Direct and the ranking controllers.

The main study uses 128 reserved episodes per task, with one standard start and two preassigned five-action prefixes. Each prefix is a recorded five-action block from the task’s training trajectories; its actions are projected onto the environment’s action Box and executed before control, keeping the goal fixed. Query sampling uses seed 26091391; the two prefix assignments use seeds 26091392 and 26091393. Ordered manifests specify the episode, initial and goal indices, environment seed, and prefix for each query. In task order (Cube, PushT, Reacher, TwoRoom), query sources are the released validation, training, validation, and validation splits. Queries use previously unused eligible episodes. For every task, all 128 query episodes are excluded in full from the retrieval memory and prefix pool. The remaining memories contain 8,000, 16,688, 8,000, and 8,000 episodes, respectively. Retrieval indices and horizon-specific statistics are computed from these memberships using the same frozen features and formula.

All tasks reuse their supplied pretrained checkpoints and original action normalizers. Retrieval memory, query episodes, and perturbation prefixes are fixed before evaluation and shared by all controllers.

The goal offset H and allowance B depend on the task (Table 5). A prefix is executed before the evaluated policy’s allowance. Policy time then starts at zero and increments by the number of executed primitives, so retrieval uses $\max(5, H - t)$. The environment clock continues through the prefix. Initial success, success during a prefix, or termination before policy entry gives every controller the corresponding common outcome. Success rates include these outcomes for all assigned queries.

A.2 Physical success

Evaluation uses the following task-specific physical predicates. They are distinct from the latent distances used for action selection.

- **Cube:** the selected object’s three-dimensional position is at most 0.04 m from its target. The single-cube data-collection environment hides the target marker; orientation and gripper position are not part of this predicate.
- **PushT:** the Euclidean norm over the concatenated agent and object position errors is strictly below 20 environment units, and the shortest circular object-angle error is below $\pi/9$. The position condition is one norm over four coordinates.
- **Reacher:** each of the two joint-position errors has absolute value below 0.05 rad. The errors use direct joint-angle differences.
- **TwoRoom:** the Euclidean distance from the two-dimensional agent position to its goal is below 16 environment units.

Success is checked initially and after every primitive. Success is determined by the task predicate at the stopping state. A controller stops on success, terminal failure, or exhaustion of its primitive-action allowance; thus execution can end before the full block. The matched construction restores the same physical state and goal before running each controller. Simulator coordinates restore matched starts and determine physical success; the controller scores encoded images.

A.3 Shared model and search implementation

The pretrained model belongs to the joint-embedding family [Assran et al., 2023, Balestriero and LeCun, 2025, Maes et al., 2026]. All controllers use the same task-specific frozen encoder. The predictive controllers also share its frozen forward model. Scoring uses FP32, ImageNet channel normalization, 224-pixel image resizing, and 192-dimensional projected latent endpoints. CUDA matrix-multiplication and cuDNN TF32 are disabled. One actual current image supplies the initial context. AP-rank pairs that current image with each of the eight action candidates in one batch.

The pretrained LeWM action StandardScaler retains its original float64 mean, scale, and affine transform, fitted to finite source-dataset action rows after removing NaNs. For every task, the scaler keeps its pretrained statistics while retrieval statistics use the active memory. For an effective raw block u , the notation $F(z, u)$ includes this original normalization. Recorded-action controllers and single-block Gaussian CEM project proposals onto the actual raw-action Box before scoring and execute the same effective raw actions. Clipping an action proposal leaves the target unchanged: it remains the encoding of the recorded successor image.

The three single-block Gaussian CEM variants use 30 iterations, 300 candidates, 30 elites, an initial normalized mean of zero, initial standard deviation $1/3$, and an elite-standard-deviation floor of 10^{-5} . The search maintains means and deviations in normalized coordinates and includes the current mean as the first candidate in each population. For scoring, candidates are clamped to the Box obtained by transforming the raw action bounds with the fixed scaler; elite means and sample standard deviations are updated from the original, unclamped proposals. The selected effective normalized block is inverse-transformed and clipped to the raw Box to remove boundary round-off. The solver compares the projected final mean with the projected initial incumbent, retaining the incumbent unless the final mean has lower cost. A common seed indexed by task and query ordinal supplies all three target variants and all start conditions. Each search starts from the fixed Gaussian prior. After the 30 populations, the final and initial means are scored together, giving 31 batches and 9,002 predicted blocks per entered decision.

Direct and the three ranking controllers use the same retrieval definition, whose eight starts can come from the same episode. AP-rank and final-goal rank share the first candidate predictions exactly; the Gaussian controllers share the first prior and search noise. Later states, retrieved records, and random-stream consumption may diverge after the policies act differently.

A.4 Aggregation and prediction work

Success rates average the two perturbed starts within each query, then queries within a task. Task means give all four tasks equal weight. Tables in this appendix show standard starts and the perturbation average; the CSV files in our repository give separate starts, sample counts, threshold sweeps, and prediction work for every setting. Original-offset results are reused across comparisons.

Prediction work counts the actual predicted five-action blocks across an episode, including early stopping. Ranking evaluates eight blocks in one batch per decision; single-block CEM evaluates $300I + 2$ blocks for I iterations. LeWM evaluates 45,000 blocks per decision using its multiblock search and executes up to 25 primitive actions before replanning. Direct uses no transition prediction for action selection. Encoding, target construction, retrieval, and interaction also contribute to elapsed runtime.

A.5 Observation-only deployment

Observation-only memory supplies ordered images, their frozen features, episode boundaries, and temporal indices. These identify the retrieved segment and the observed target. CEM generates candidate actions and passes them to the frozen predictor, which was pretrained with actions. Both memory settings use the same retrieval and target construction.

Algorithm 1 AP-rank: an observed objective for a frozen predictor

Require: frozen E, F ; memory $\{\mathcal{B}_h\}$; current image o_0 ; goal image o_g ; supplied H ; action allowance B

```
1:  $L \leftarrow 5; k \leftarrow 8; z_g \leftarrow E(o_g); t \leftarrow 0$ 
2: while  $t < B$ , the goal is not reached, and the environment is active do
3:    $z_t \leftarrow E(o_t); h \leftarrow \max\{L, H - t\}$ 
4:   retrieve  $(s_1, \dots, s_k)$  from  $\mathcal{B}_h$  using Equation (5)
5:    $q_{\text{obs}} \leftarrow E(o_{s_1+L}); \mathcal{C} \leftarrow \{P(u_{s_1}), \dots, P(u_{s_k})\}$ 
6:   predict  $\{F(z_t, u) : u \in \mathcal{C}\}$  in one batch
7:   select  $\hat{u}$  by Equation (7), breaking ties by retrieval order
8:   execute up to  $L$  primitives from  $\hat{u}$ , stopping at success, termination, or  $B$ 
9:   increase  $t$  by the executed primitives; observe the next image  $o_t$ 
10: end while
```

A.6 Algorithm and implementation details

Algorithm 1 gives the AP-rank procedure used in every main-study task and condition. Section 3 defines how it retrieves a record, sets a target, selects a clipped action block, and replans after execution. The preceding protocol defines the paired starts and physical success criteria.

Single-block CEM and recorded-action controllers use raw-action Box projection. LeWM retains its released action mapping and execution settings (Appendix D). The independent supporting study uses the mapping specified in Appendix F.

The pretrained networks and original action normalizer stay fixed. Live-query features for retrieval use CUDA bfloat16 autocast and are returned in float32; they are compared against the preserved frozen-cache features. Current and target encodings for action scoring are computed separately in FP32. Retrieval-key statistics are computed from the records in the active horizon bank. Coordinate-wise population standard deviations are floored at 10^{-4} , and the source, distant endpoint, and displacement parts have equal weights. Bank membership and these statistics stay fixed; the eight closest starts may share an episode. Once encoded, a fixed target can be reused across candidate evaluations. Each prediction starts with a single current image. LeWM subsequently accumulates up to three context blocks within its rollout.

The experiments use H20 GPUs. The main study uses Python 3.11.14, PyTorch 2.7.1 with CUDA 12.6 and cuDNN 90501. MuJoCo is 3.12.0 for Cube, PushT, and TwoRoom; Reacher uses 3.10.0 with dm-control 1.0.43.

B Properties of Target-Based Selection

B.1 A continuous counterexample with smooth dynamics

Proof of Proposition 1. Let $\mathcal{X} = \{c(s) : s \in \mathbb{R}\}$, where $c(s) = (s, s^2)$, and use the identity encoder. A primitive action $a \in [0, 1/8]$ has transition

$$T((x, y), a) = (x + a, y + 2xa + a^2).$$

This polynomial map is the unit-time flow of $\dot{x} = a, \dot{y} = 2xa$ under constant control and preserves $\mathcal{X}$. Thus a five-action block $u \in \mathcal{C} = [0, 1/8]^5$ has exact endpoint

$$F(c(s), u) = c\left(s + \sum_{i=1}^5 a_i\right).$$

Both controllers globally minimize squared endpoint distance over the entire continuous set $\mathcal{C}$, execute the chosen block and observe again. Only the target differs.

For the final goal $g = c(2)$, write

$$D(s) = \|c(s) - g\|^2 = (s - 2)^2 + (s^2 - 4)^2, \quad D'(s) = 2(s - 2)(2s^2 + 4s + 1).$$

The quadratic roots are $-1 \pm 1/\sqrt{2}$, so $D'(s) > 0$ throughout $[-3/2, -3/8]$. From any $s_0 \in [-3/2, -1]$, an admissible block ends at $c(s_0 + d_u)$ with $0 \leq d_u \leq 5/8$, within that increasing region. Every $d_u > 0$ increases final-goal distance. Nonnegative actions make the zero block the unique global minimizer. It leaves the state unchanged, so final-goal selection never reaches g .

Now take one feasible recorded trajectory with consecutive block endpoints

$$(r_0, r_1, r_2, r_3, r_4, r_5, r_6) = \left(-\frac{3}{2}, -\frac{7}{8}, -\frac{1}{4}, \frac{3}{8}, 1, \frac{13}{8}, 2\right).$$

Its first five blocks use $a_i = 1/8$ and the last uses $a_i = 3/40$. Supply the observed targets $c(r_1), \dots, c(r_6)$ in order. For every allowed s_0 , the first increment $r_1 - s_0 \in [1/8, 5/8]$ is feasible. Each later increment is also positive and at most $5/8$. At each decision a block attains zero target cost, so every global minimizer reaches that target exactly. Induction gives the final goal in six blocks. Ties between primitive sequences do not change their endpoint; primitive-level stopping can only shorten the final block. $\square$

The dynamics admit a continuous route from every stated start. Failure under final-goal scoring comes from rejecting the temporary increase in distance needed to follow it. The prescribed recorded sequence isolates the target's role: it turns the same exact dynamics and local optimizer into a goal-reaching controller.

The failure persists with symmetric actions. Allow $a_i \in [-1/8, 1/8]$ and write $\alpha = -1 - 1/\sqrt{2}$, $\beta = -1 + 1/\sqrt{2}$. Below β , D decreases to α and then increases. From $s_0 \in [-3/2, -1]$, the reachable interval $[s_0 - 5/8, s_0 + 5/8]$ lies below β . Its minimizing coordinate is $\max\{\alpha, s_0 - 5/8\}$; repeating this rule reaches α within two blocks. Since $\alpha + 5/8 < \beta$, subsequent minimizing blocks also end at α . The same observed-target sequence remains feasible from each original start and reaches the goal. Allowing both action directions thus preserves failure under final-goal scoring.

B.2 How endpoint errors affect action selection

At a fixed live state and target q , let e_u be an action block's exact encoded endpoint and $r_u = F(z_t, u) - e_u$ its prediction error. The resulting error in target cost is

$$\xi_u(q) = J_q(u) - \|e_u - q\|^2 = 2(e_u - q)^\top r_u + \|r_u\|^2.$$

The predicted cost difference between u and v therefore differs from the exact one by $\xi_u(q) - \xi_v(q)$. Two endpoint errors with the same size can change action costs differently: the dot product depends on the direction of each error relative to the target. If both costs change by the same amount, their ordering stays unchanged.

If exact target costs favor u over v by a gap $m > 0$, their preference is preserved whenever $\xi_u(q) - \xi_v(q) < m$. This explains why endpoint error and selection regret provide complementary information in Table 4: the first measures endpoint accuracy, while the second measures the cost of the action chosen under the target.

Proposition 1 concerns the target itself. With $r_u = 0$, every predicted target cost is exact, yet final-goal scoring still favors staying still in the construction. Changing the target changes those costs without changing the dynamics.

C RQ1: Target Choice and Control

The main target comparisons use the shared protocol in Appendix A. This section gives target-model selection and the physical behavior behind the success rates in Table 1.

C.1 Learned intermediate targets

We train a target predictor on the same memory episodes available to Anchored Planning. For each task, a fixed episode split reserves five percent of memory for validation. Control-query episodes are excluded from both splits. Each training example samples an offset h uniformly from $5, \dots, H_{\text{task}}$, where H_{task} is the task’s main goal offset, and then a valid source position s uniformly among the training positions at that offset. The input is $(z_s, z_{s+h}, z_{s+h} - z_s, h/H_{\text{task}})$, and the target is the observed successor z_{s+5} . All encodings come from the frozen LeWM encoder in FP32.

The predictor is an MLP with three hidden layers, each using LayerNorm and GELU. We compare widths 512 and 1024, with either an absolute output or a residual added to z_s . Inputs and outputs are standardized coordinate-wise using 262,144 sampled training transitions. AdamW uses batch size 1024, learning rate 3×10^{-4} , weight decay 10^{-4} , and cosine decay over at most 50,000 updates. Validation uses the same 4096 held-out transitions for all four configurations. We evaluate every 500 updates and stop after ten checks without improvement, once at least 5000 updates have run. The selected checkpoint minimizes unstandardized latent-coordinate MSE, which gives absolute and residual models the same selection scale.

At execution, the MLP receives the current encoding, final-goal encoding, their difference, and $h(t)/H_{\text{task}}$. Its output replaces the scoring target for either Gaussian CEM or ranking the retrieved action blocks. Gaussian CEM retains its initialization and search settings; ranking scores the same retrieved blocks. Both retain the frozen predictor, action projection, and execution rule of the main study.

Table 6: **Selecting learned targets on held-out memory.** Validation MSE averages squared error over latent coordinates; bold marks the selected configuration.

Task	Width	Output	Best step	Validation MSE
Cube	512	Absolute	48,500	0.104926
	512	Residual	48,500	0.104579
	1024	Absolute	41,000	0.104422
	1024	Residual	48,500	0.102646
PushT	512	Absolute	48,000	0.023950
	512	Residual	48,000	0.024007
	1024	Absolute	47,500	0.020210
	1024	Residual	45,500	0.019658
Reacher	512	Absolute	11,500	0.391840
	512	Residual	21,500	0.389188
	1024	Absolute	11,500	0.391255
	1024	Residual	21,500	0.389396
TwoRoom	512	Absolute	14,500	0.697172
	512	Residual	15,500	0.702978
	1024	Absolute	8,000	0.697239
	1024	Residual	7,500	0.704444

Table 2 relates target accuracy to control. Target accuracy uses 4096 held-out transitions per task and excludes each source episode from retrieval. Successor errors and nearest-memory distances are squared latent distances. Our repository gives their means and medians in `target_quality.csv`; an observed target has zero distance to memory by construction.

C.2 Physical behavior along a rollout

Physical error is measured in units of the success thresholds. For Cube and TwoRoom, $\hat{e}_t$ is position error divided by 0.04 m and 16 units. For Reacher, it is the larger joint error divided by 0.05 rad. For PushT, it is the maximum of combined position error divided by 20 and wrapped block-angle error divided by $\pi/9$. Success retains the boundary conventions in Appendix A.

A failed rollout stalls when its task-relevant coordinates remain within one quarter of their success thresholds from the state at the beginning of the last $W = 10$ decisions. Shorter rollouts use their available window. A successful rollout makes a detour when $\hat{e}_t$ rises by at least $\delta = 0.5$ above its previous running minimum.

Table 7 gives the rates and eligible counts. Stalling conditions on failures with decisions; detours condition on policy-entered successes. Perturbed behavior pools eligible rollouts from both prefixes. Table 8 varies the window and threshold for the three CEM target rules.

Table 7: **Physical behavior at the original goal offsets.** Percentages are followed by positive/eligible rollout counts. Stalling uses $W = 10$ and conditions on failures with decisions; detours use $\delta = 0.5$ and condition on policy-entered successes. Perturbed pools eligible rollouts from both prefixes.

Task	Controller	Standard		Perturbed	
		Stalling	Detour	Stalling	Detour
Cube	LeWM	39.8 (43/108)	10.5 (2/19)	48.2 (108/224)	6.7 (2/30)
	CEM (final goal)	100.0 (119/119)	37.5 (3/8)	100.0 (249/249)	40.0 (2/5)
	CEM (learned)	79.2 (38/48)	60.8 (48/79)	89.2 (140/157)	68.0 (66/97)
	AP-CEM	86.8 (59/68)	66.1 (39/59)	92.2 (166/180)	73.0 (54/74)
	CEM (transported)	91.3 (73/80)	68.1 (32/47)	95.4 (186/195)	71.2 (42/59)
	Direct	87.0 (20/23)	60.6 (63/104)	94.0 (126/134)	64.2 (77/120)
	Final-goal rank	93.8 (45/48)	55.7 (44/79)	99.3 (148/149)	61.0 (64/105)
	Learned-target rank	100.0 (11/11)	64.7 (75/116)	89.1 (114/128)	63.5 (80/126)
	AP-rank	100.0 (34/34)	61.3 (57/93)	92.8 (129/139)	67.0 (77/115)
PushT	LeWM	28.6 (34/119)	88.9 (8/9)	28.0 (65/232)	83.3 (20/24)
	CEM (final goal)	80.0 (100/125)	33.3 (1/3)	78.7 (196/249)	42.9 (3/7)
	CEM (learned)	80.3 (49/61)	98.5 (66/67)	81.7 (107/131)	99.2 (124/125)
	AP-CEM	92.3 (36/39)	100.0 (89/89)	83.3 (75/90)	100.0 (166/166)
	CEM (transported)	87.8 (43/49)	100.0 (79/79)	78.2 (111/142)	100.0 (114/114)
	Direct	95.2 (59/62)	100.0 (66/66)	91.0 (191/210)	100.0 (46/46)
	Final-goal rank	90.5 (86/95)	100.0 (33/33)	88.3 (197/223)	100.0 (33/33)
	Learned-target rank	93.1 (54/58)	100.0 (70/70)	87.3 (144/165)	98.9 (90/91)
	AP-rank	90.7 (39/43)	100.0 (85/85)	89.6 (120/134)	100.0 (122/122)
Reacher	LeWM	0.0 (0/43)	87.1 (74/85)	0.0 (0/80)	90.8 (157/173)
	CEM (final goal)	0.0 (0/95)	51.5 (17/33)	0.0 (0/198)	47.3 (26/55)
	CEM (learned)	0.0 (0/58)	92.9 (65/70)	0.0 (0/105)	93.9 (139/148)
	AP-CEM	0.0 (0/31)	96.9 (94/97)	0.0 (0/54)	97.5 (194/199)
	CEM (transported)	0.0 (0/24)	96.2 (100/104)	0.0 (0/50)	97.5 (198/203)
	Direct	0.0 (0/1)	97.6 (124/127)	0.0 (0/8)	99.2 (243/245)
	Final-goal rank	0.0 (0/12)	94.0 (109/116)	0.0 (0/32)	95.0 (210/221)
	Learned-target rank	0.0 (0/6)	100.0 (122/122)	0.0 (0/8)	99.6 (244/245)
	AP-rank	0.0 (0/4)	98.4 (122/124)	0.0 (0/11)	99.6 (241/242)
TwoRoom	LeWM	0.0 (0/106)	68.2 (15/22)	0.0 (0/213)	51.2 (22/43)
	CEM (final goal)	33.3 (42/126)	0.0 (0/2)	32.1 (81/252)	0.0 (0/4)
	CEM (learned)	0.0 (0/60)	27.9 (19/68)	0.0 (0/115)	27.0 (38/141)
	AP-CEM	19.1 (13/68)	73.3 (44/60)	15.4 (21/136)	75.8 (91/120)
	CEM (transported)	41.7 (35/84)	59.1 (26/44)	39.8 (66/166)	64.4 (58/90)
	Direct	0.0 (0/1)	96.9 (123/127)	– (0/0)	99.2 (254/256)
	Final-goal rank	7.1 (1/14)	93.9 (107/114)	6.9 (2/29)	96.5 (219/227)
	Learned-target rank	25.0 (1/4)	97.6 (121/124)	0.0 (0/4)	96.8 (244/252)
	AP-rank	0.0 (0/2)	98.4 (124/126)	– (0/0)	99.2 (254/256)

Table 8: **Behavior across thresholds.** Rates (%) weight the four tasks equally, using each task’s eligible rollout counts. Our repository gives task-level counts at every threshold.

Controller	Start	Stalling window			Detour threshold		
		$W = 5$	10	20	$\delta = 0.25$	0.5	1.0
CEM (final goal)	Standard	55.1	53.3	49.9	68.6	30.6	29.1
CEM (final goal)	Perturbed	54.9	52.7	49.7	46.4	32.5	19.0
CEM (learned)	Standard	46.1	39.9	31.4	77.6	70.0	53.7
CEM (learned)	Perturbed	45.6	42.7	32.1	79.3	72.0	58.9
AP-CEM	Standard	51.7	49.5	40.5	90.6	84.1	70.7
AP-CEM	Perturbed	52.2	47.7	39.3	91.5	86.6	71.8

C.3 A matched PushT rollout

Figure 6 shows the first main-study query in the fixed ordering for which AP-rank succeeds and Direct fails, testing the first prefix before the second. It plots logged physical states throughout execution.

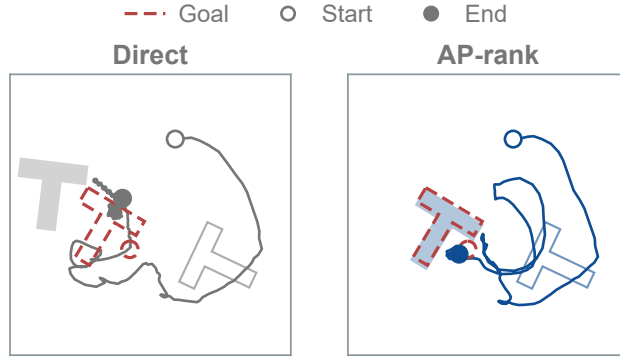


Figure 6: **Predictive ranking changes a matched PushT execution.** Both controllers share the perturbed start and goal. Curves trace the agent; circles and T shapes depict the agent and pushed block. Hollow and filled shapes show initial and terminal poses; dashed outlines mark the goal.

D RQ2: Search Budget and Goal Distance

D.1 Search budgets

The budget sweep changes CEM iterations over $\{1, 2, 5, 10, 30\}$ while preserving the main queries, perturbation prefixes, frozen models, and raw-action Box projection. Each budget starts with the same query seed and initial proposals; later random draws follow its iteration count. The largest-budget points reuse the main outcomes. Table 9 gives both start conditions.

Table 9: **Success across search budgets.** Success (%) uses the main queries at the original offsets. Only CEM iterations change; each decision evaluates $300I + 2$ predicted blocks. Perturbed averages both starts within query.

Task	Iterations	Final goal		Learned target		Observed target	
		Standard	Perturbed	Standard	Perturbed	Standard	Perturbed
Cube	1	3.9	2.0	11.7	6.6	10.9	7.0
	2	3.1	2.3	40.6	26.2	26.6	15.6
	5	3.1	2.3	52.3	34.0	36.7	26.6
	10	4.7	3.9	59.4	35.2	44.5	29.7
	30	7.0	2.7	62.5	38.7	46.9	29.7
PushT	1	1.6	2.0	3.9	3.9	3.1	1.6
	2	1.6	2.0	25.8	27.0	8.6	10.9
	5	1.6	2.0	44.5	41.8	43.8	43.4
	10	1.6	2.7	48.4	44.5	69.5	68.0
	30	2.3	2.7	52.3	48.8	69.5	64.8
Reacher	1	16.4	15.2	48.4	50.0	59.4	56.6
	2	18.8	18.4	50.8	53.9	64.8	67.2
	5	24.2	20.3	60.2	59.8	78.9	75.8
	10	23.4	23.4	63.3	60.5	85.9	75.4
	30	25.8	22.7	54.7	59.0	75.8	78.9
TwoRoom	1	0.0	0.0	34.4	35.2	7.0	9.0
	2	0.0	0.0	39.1	41.0	19.5	20.3
	5	0.8	0.0	46.9	49.2	37.5	37.1
	10	0.8	1.6	55.5	53.5	43.0	45.3
	30	1.6	1.6	53.1	55.1	46.9	46.9

D.2 LeWM configuration

LeWM uses the released `WorldModelPolicy` and `CEMSolver` from `stable-worldmodel 0.0.6`. Its five model blocks each contain five primitive actions; the policy executes all five blocks before replanning, stopping earlier on success, termination, or the action allowance. CEM uses 300 candidates, 30 iterations, 30 elites, initial standard deviation 1, and the released history and mean updates. Actions pass through the pretrained inverse scalar directly to the environment. Starts, goals, action allowances, checkpoints, and success predicates match the target comparisons. The single-block CEM variants retain their shared search and change only the target.

Released evaluation protocol. The released LeWM evaluator samples valid rows from the supplied datasets, takes the goal image 24 actions after the start, and allows 50 actions. Our $H = 25$ queries place the goal 25 actions after the start and allow 25 actions on Cube and 50 on the other tasks. The released evaluator scores the dataset image of the start, whereas we render the restored start; on Cube, it restores the full recorded `qpos/qvel`, whereas every controller in our study starts from a state reconstructed from the public observation. Its configuration file uses 30 CEM iterations on every task, whereas [Maes et al. \[2026\]](#) report ten outside PushT.

Table 10 lists the published rates next to runs of the released evaluator and of our implementation on the same queries, action mapping, and CEM random streams. The two implementations give the same outcome on 599 of 600 episodes.

Table 10: **LeWM under its released evaluation protocol.** Published success (%) from Figure 6 of [Maes et al. \[2026\]](#), and success measured with the released evaluator and with our implementation on the same queries. Measured values average seeds 42–44 with 50 queries each; the released configuration uses seed 42.

Task	Paper report	Released evaluator	Our implementation
Cube	74.0	74.7	74.7
PushT	96.0	92.0	91.3
Reacher	86.0	80.0	80.0
TwoRoom	87.0	87.3	87.3

On Cube and PushT, query sampling produces the largest change among the factors we varied one at a time: with the released sampling rule, LeWM success in our $H = 25$ setting rises from 32.0% to 71.9% and from 60.9% to 82.8%, whereas a 50-action allowance alone raises Cube to 36.7%. Rescoring 512 saved main-study episodes with the released success predicates changes no outcome. The `lewm_checks` folder in our repository lists query identities, per-seed outcomes, and each single-factor result, including the goal frame, initial image, and Cube initialization.

Five-action replanning. The additional variant replans after at most five primitive actions instead of 25. It retains the released lookahead, candidate count, iterations, elites, initial deviation and warm start, and uses the main queries, goals and action allowances (Table 11). Our repository gives complete outcomes and episode prediction work in `lewm_checks/lewm_replan5.csv` and `lewm_replan5_episodes.csv`.

Table 11: **LeWM with five-action replanning.** The variant retains 25-action lookahead and the released CEM settings, changing only execution to one five-action block. Results use the main queries and original goal offsets: 128 standard starts and 256 perturbed starts per task. Work counts mean predicted five-action blocks per episode, including early stopping.

Task	Success (%)		Work (10^6 blocks)	
	Standard	Perturbed	Standard	Perturbed
Cube	13.3	13.3	1.189	1.197
PushT	7.0	8.2	2.402	2.374
Reacher	73.4	77.3	1.388	1.322
TwoRoom	27.3	25.8	1.500	1.509

D.3 Recorded goal offsets

Each goal is the observation at the selected offset from the original query source. Cube, PushT, and Reacher add $H = 25, 50, 100$; TwoRoom adds $H = 25, 50$. The action allowance preserves each task’s original B/H . The learned target retains its training normalization H_{task} . Original-offset points reuse the main outcomes. Table 12 gives synthesis and ranking results.

D.4 Prediction work

Table 13 collects mean episode prediction counts at the original offsets, including early stopping. A count represents one predicted five-action block. LeWM can execute up to 25 actions between decisions, whereas the other predictive controllers execute up to five. Encoding, target construction, retrieval, and interaction also contribute to elapsed runtime. Our repository reports prediction counts for every budget, goal offset, and ablation setting.

Table 12: **Success across recorded goal offsets.** Success (%) uses the same query sources. The goal moves to the observation H actions later, with each task’s B/H fixed. Prediction lookahead and target-model training remain unchanged. Std. and Pert. denote standard starts and the query-wise perturbation average.

Action synthesis

Task	H	LeWM		Final goal		Learned		AP-CEM	
		Std.	Pert.	Std.	Pert.	Std.	Pert.	Std.	Pert.
Cube	25	32.0	33.6	28.9	26.2	75.0	53.9	63.3	49.6
	50	16.4	12.1	7.0	4.3	58.6	37.9	47.7	30.5
	100	18.8	13.3	8.6	9.0	55.5	34.8	32.0	25.4
	150	15.6	12.5	7.0	2.7	62.5	38.7	46.9	29.7
PushT	25	60.9	57.8	51.6	46.9	91.4	75.4	91.4	87.5
	50	22.7	21.9	6.3	9.0	76.6	63.7	82.0	81.6
	100	10.9	9.4	3.9	4.3	57.0	48.4	77.3	73.8
	140	7.0	9.4	2.3	2.7	52.3	48.8	69.5	64.8
Reacher	25	68.8	67.6	66.4	66.0	57.8	57.8	67.2	71.1
	50	95.3	92.6	57.0	52.3	53.1	53.5	81.3	76.2
	100	83.6	82.0	35.2	35.2	48.4	52.3	84.4	76.6
	150	66.4	68.8	25.8	22.7	54.7	59.0	75.8	78.9
TwoRoom	25	77.3	74.2	30.5	34.8	99.2	99.2	89.1	85.9
	50	36.7	35.5	5.5	6.6	94.5	94.1	72.7	73.8
	100	17.2	16.8	1.6	1.6	53.1	55.1	46.9	46.9

Recorded-action selection

Task	H	Final goal		Learned		AP-rank	
		Std.	Pert.	Std.	Pert.	Std.	Pert.
Cube	25	92.2	60.9	88.3	57.8	92.2	59.0
	50	68.0	43.8	83.6	48.4	70.3	44.1
	100	59.4	34.8	78.9	43.8	64.8	37.1
	150	62.5	41.8	91.4	50.0	73.4	45.7
PushT	25	89.8	68.0	96.1	68.4	95.3	71.1
	50	67.2	36.7	81.3	48.4	84.4	55.9
	100	26.6	14.1	64.1	38.7	71.1	46.9
	140	25.8	12.9	54.7	35.5	66.4	47.7
Reacher	25	92.2	93.0	87.5	88.3	91.4	91.8
	50	97.7	97.7	94.5	93.8	97.7	96.5
	100	95.3	94.9	97.7	96.9	97.7	96.1
	150	90.6	87.5	95.3	96.9	96.9	95.7
TwoRoom	25	100.0	99.2	100.0	99.6	100.0	100.0
	50	94.5	94.1	100.0	99.6	100.0	100.0
	100	89.1	88.7	96.9	98.4	98.4	100.0

Table 13: **Mean episode prediction work at the original offsets.** Values are thousands of predicted five-action blocks and include early stopping. Std. denotes standard starts and Pert. the query-wise perturbation average. Setting-specific work for every sweep is provided in `interventions.csv`.

Controller	Cube		PushT		Reacher		TwoRoom	
	Std.	Pert.	Std.	Pert.	Std.	Pert.	Std.	Pert.
LeWM	235.20	241.88	514.69	505.37	255.59	244.51	320.98	321.50
CEM (final goal)	252.62	263.17	493.56	491.10	407.76	425.31	354.95	354.95
CEM (learned)	142.27	191.54	375.06	387.40	393.91	379.98	250.65	247.94
AP-CEM	170.26	206.84	326.11	338.63	323.58	313.17	262.61	265.07
CEM (transported)	193.40	223.26	347.84	393.80	306.56	310.50	289.19	291.37
Direct	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Final-goal rank	0.12	0.16	0.38	0.42	0.20	0.20	0.15	0.16
Learned-target rank	0.08	0.15	0.33	0.37	0.24	0.23	0.14	0.14
AP-rank	0.10	0.15	0.30	0.34	0.24	0.24	0.14	0.14

E RQ3: Memory Targets and Predictive Selection

E.1 Target offsets, memory size, and retrieval keys

The target-offset ablation uses $q = z_{s_1+\ell}$ with $\ell \in \{1, 3, 10\}$ while prediction and execution retain $L = 5$. The default uses $\ell = L$; retrieval keeps only sources with a valid successor at the chosen offset. Memory-size ablations use nested episode subsets of 10, 25, and 50 percent, sampled with seed 26091600, and recompute retrieval standard deviations in each subset. Table 14 gives task-level results for both action rules.

Key ablations omit either the distant endpoint or the displacement component. The fixed-span ablation uses $h = H$ throughout execution. Other retrieval and control settings remain unchanged (Table 15).

Table 14: **Target offsets and memory size.** Success (%) uses the main queries. The default observes the five-step successor and uses the full memory. Prediction and execution retain five-action blocks throughout.

Setting	Cube		PushT		Reacher		TwoRoom	
	Std.	Pert.	Std.	Pert.	Std.	Pert.	Std.	Pert.
<i>AP-CEM</i>								
Default	46.9	29.7	69.5	64.8	75.8	78.9	46.9	46.9
$\ell = 1$	43.8	31.3	34.4	29.3	50.0	44.9	43.8	46.1
$\ell = 3$	61.7	42.6	68.8	60.9	91.4	90.6	57.8	56.3
$\ell = 10$	19.5	15.6	58.6	57.4	87.5	89.8	50.8	48.0
10% memory	39.1	25.4	59.4	58.6	75.8	76.2	52.3	48.4
25% memory	39.1	27.7	64.8	63.3	79.7	76.2	50.0	46.9
50% memory	43.8	30.1	67.2	61.3	76.6	76.2	46.9	47.7
<i>AP-rank</i>								
Default	73.4	45.7	66.4	47.7	96.9	95.7	98.4	100.0
$\ell = 1$	71.9	44.5	30.5	21.9	93.8	94.9	82.0	84.0
$\ell = 3$	71.1	44.9	49.2	35.5	95.3	96.1	98.4	97.3
$\ell = 10$	71.1	44.1	59.4	34.4	96.9	95.7	99.2	99.2
10% memory	63.3	42.2	53.1	39.5	94.5	94.5	99.2	99.6
25% memory	67.2	44.1	58.6	43.0	96.1	95.7	98.4	98.8
50% memory	73.4	45.3	61.7	43.4	93.8	95.7	99.2	99.6

Table 15: **Retrieval keys and temporal span.** AP-rank success (%) uses the main queries. Omitting a key component leaves the other components unchanged; fixing $h = H$ replaces the adaptive span $\max(5, H - t)$.

Setting	Cube		PushT		Reacher		TwoRoom	
	Std.	Pert.	Std.	Pert.	Std.	Pert.	Std.	Pert.
Full key	73.4	45.7	66.4	47.7	96.9	95.7	98.4	100.0
Without endpoint	81.3	46.5	67.2	50.4	94.5	94.5	100.0	100.0
Without displacement	73.4	45.7	63.3	44.9	96.1	97.7	99.2	100.0
Fixed span $h = H$	4.7	3.1	0.0	0.0	12.5	19.1	21.1	23.0

E.2 Absolute endpoints and transported displacements

The anchoring comparison uses the observation-only Gaussian controller on the main queries. For a live encoding z_t , retrieved source z_s , and its successor observation z_{s+L} , the two target rules are

$$q_{\text{obs}} = z_{s+L}, \quad q_{\text{transport}} = z_t + z_{s+L} - z_s, \quad L = 5. \quad (9)$$

The live, source, and successor images use the same frozen encoder in FP32. Both rules construct targets from ordered observations; the common Gaussian prior supplies actions. Their difference is $q_{\text{obs}} - q_{\text{transport}} = z_s - z_t$: one pursues the recorded endpoint, while the other matches the recorded displacement from the live encoding. The targets coincide when the source and live encodings coincide.

How the target changes action ordering. Suppose the current encoding has already moved partway along the recorded displacement. Anchoring requests only the remaining movement to the recorded endpoint. Transport repeats the full displacement from the new state, placing its target beyond that endpoint.

Let $e = z_t - z_s$ denote the current offset from the recorded source. Applying Equation (2) to the two targets gives

$$\Delta_{q_{\text{obs}}}(u, v) - \Delta_{q_{\text{transport}}}(u, v) = 2(p_u - p_v)^\top e. \quad (10)$$

After a shift, the dot product in Equation (10) gives the change in the two actions' cost difference. Its sign determines which action gains relative to the other. Perturbed starts change the live state on which both rules act; within each matched start, the target variants share their initial retrieved record.

Table 16: **Anchoring and displacement transport on the main queries.** Success (%) compares the observed endpoint with transported motion under the same Gaussian search. Perturbed averages both starts within query; the mean weights tasks equally.

Task	Start	Observed target	Transported target
Cube	Standard	46.9	37.5
	Perturbed	29.7	23.8
PushT	Standard	69.5	61.7
	Perturbed	64.8	44.5
Reacher	Standard	75.8	81.3
	Perturbed	78.9	80.5
TwoRoom	Standard	46.9	34.4
	Perturbed	46.9	35.2
Mean	Standard	59.8	53.7
	Perturbed	55.1	46.0

E.3 Exact endpoints and the first action block

At each assigned start, the simulator executes the eight retrieved action blocks from the same saved physical state. The frozen encoder maps their resulting observations to exact candidate endpoints. Endpoint-error and selection-regret means include starts for which all eight candidates execute five actions. Endpoint error first averages over candidates. For perturbed starts, each query averages its eligible starts, after which queries receive equal weight. Eligible-start and query counts are reported in `diagnostics.csv`.

The first-block intervention compares Direct, the frozen predictor, simulator endpoints, and transported displacement estimates. Predictor and simulator endpoints are each scored against the final, observed, and learned targets; displacement uses the observed target. After the selected first block, every rollout uses Direct. Success rates retain all 128 assigned queries per task, including starts resolved before a policy decision.

Table 4 reports endpoint error and regret for standard starts and the perturbation average. Table 17 gives every first-block selector with its common continuation. Separate-prefix measurements are included in `diagnostics.csv` and `first_block.csv`.

Table 17: **Changing only the first action block.** Success (%) follows the selected first block with the same Direct continuation. Predictor and simulator selectors score the same eight candidates against the indicated target. All 128 assigned queries per task are included.

First-block selector	Cube		PushT		Reacher		TwoRoom	
	Std.	Pert.	Std.	Pert.	Std.	Pert.	Std.	Pert.
Direct	82.0	47.7	51.6	18.0	99.2	96.9	99.2	100.0
Predictor / Final	67.2	43.0	39.1	18.4	96.1	97.3	100.0	100.0
Simulator / Final	73.4	44.5	43.0	19.1	96.1	97.3	99.2	99.6
Predictor / Learned	85.9	48.8	45.3	18.8	96.9	97.7	100.0	99.6
Simulator / Learned	86.7	49.2	42.2	22.3	96.9	97.7	100.0	99.6
Predictor / Observed	78.9	47.3	53.1	22.3	95.3	96.1	99.2	99.2
Simulator / Observed	80.5	48.8	53.9	23.8	96.9	96.1	100.0	99.2
Displacement / Observed	78.1	48.4	50.8	23.0	98.4	95.7	99.2	99.6

F Independent-Query Target Replication

The supporting study uses 128 validation episodes per task, disjoint from the main queries, with the same task-specific goal offsets, action allowances, frozen models, and physical success criteria. It passes de-normalized actions directly to the environment, without the external raw-action Box projection used by the main single-block controllers. Each target contrast shares this query set and action mapping.

Gaussian CEM compares final-goal and observed targets at standard starts, using 30 iterations, 300 candidates, 30 elites, and initial standard deviation $1/3$. Ranking compares the same targets on eight retrieved action blocks at the standard start and two preassigned five-action perturbations. The goal and memory remain fixed across starts; all assigned outcomes enter success rates. Table 18 collects the comparisons.

Table 18: **Target effects on independent supporting queries.** Success (%) compares final-goal and observed targets within each action rule. Ranking includes standard and perturbed starts; CEM was evaluated at standard starts. Perturbed averages the two assigned starts within query.

Task	CEM: Standard		Ranking: Standard		Ranking: Perturbed	
	Final	Observed	Final	Observed	Final	Observed
Cube	8.6	39.1	64.8	71.1	41.4	43.4
PushT	1.6	81.3	25.8	78.1	9.0	49.2
Reacher	25.0	75.8	92.2	95.3	88.7	95.7
TwoRoom	0.8	39.1	85.2	100.0	88.3	99.6

Observed targets improve success over final-goal scoring in every task under both action rules. Our repository gives results for each perturbation separately; Appendix E reports the main-query counterparts.